

MODULE 36 · DOMAIN 4 - ACCELERATE WORKLOAD MIGRATION AND MODERNIZATION

Data Migration

Over the wire or in a box. The arithmetic decides, not the instinct.

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An honest warning, again

- ◆ **Snowmobile retired in 2024. Snowcone was discontinued in 2024**, with support ending in 2025.
- ◆ **Snowball Edge closed to new customers on 7 November 2025**, the same day as Migration Hub and Application Discovery Service.
- ◆ **AWS will discontinue Snowball support in all commercial Regions on 31 December 2026**. This one has an end date.
- ◆ **The Snow Family remains on the exam's in-scope list**, so it remains the examinable answer. Answer with it.

ON THE EXAM

AWS points new customers at DataSync, AWS Data Transfer Terminal, partner appliances, and Outposts for edge compute. None is in scope.

Divide the data by the bandwidth. **If the answer is longer than the deadline, no amount of tooling helps.**

The arithmetic, done

LINK	PER DAY, AT 100%	500 TB TAKES
100 Mbps	About 1 TB	Well over a year
1 Gbps	About 10.8 TB	About 46 days, perfect
10 Gbps	About 108 TB	About 5 days, perfect
10 Gbps at 60%	About 65 TB	About 8 days, realistic

ON THE EXAM

Always discount for real utilisation. A link that is also carrying production traffic is not available to you at 100%.

AWS DataSync

- ◆ **Online transfer between file systems and object stores**, on-premises to AWS, AWS to AWS, or from other clouds.
- ◆ **Handles the hard parts:** integrity validation, retries, scheduling, incremental sync and bandwidth throttling.
- ◆ Sources include NFS, SMB, HDFS and object storage. Targets include S3, EFS, FSx and other Regions.
- ◆ **Incremental after the first run**, which is what makes it a synchronisation tool rather than a one-off copy.

ON THE EXAM

This is what people rebuild badly with rsync and cron. Validation, retry and throttling are the parts a script always lacks.

The Snow Family, as examinable

- ◆ **Snowball Edge Storage Optimized:** 210 TB of NVMe, transferring up to 1.5 GB/s locally. The data migration device.
- ◆ **Snowball Edge Compute Optimized:** 104 vCPUs and 28 TB, for processing at the edge rather than bulk transfer.
- ◆ Order, receive, copy locally, ship back, AWS imports to S3. The delay is shipping, and it is predictable.
- ◆ Cluster several devices for larger datasets. Snowmobile, which handled 100 PB, no longer exists.

ON THE EXAM

The exam still tests this. A remote site with poor connectivity and a hard deadline is the classic Snowball scenario.

Transfer Family and Transfer Acceleration

- ◆ **AWS Transfer Family** gives managed SFTP, FTPS, FTP and AS2 endpoints backed by S3 or EFS.
- ◆ Its purpose is **ongoing exchange with third parties** who will not change their process, not bulk migration.
- ◆ **S3 Transfer Acceleration** routes uploads over the CloudFront edge network onto the AWS backbone.
- ◆ Acceleration helps distant clients over long distances. It does nothing for a client already close to the Region.

ON THE EXAM

Transfer Family replaces an SFTP server you were operating. It is a steady-state integration, not a migration tool.

Reading the scenario

THE SCENARIO SAYS

Petabytes, poor connectivity, hard deadline

Ongoing sync from on-premises NFS

Partners upload by SFTP and will not change

Global users uploading to one bucket, slowly

Copy between AWS Regions or services

A script with rsync and cron

IT IS POINTING AT

Snow Family. Do the arithmetic first

DataSync, incremental after first run

AWS Transfer Family

S3 Transfer Acceleration

DataSync also does AWS to AWS

DataSync: validation, retry, throttling

ON THE EXAM

Do the division before choosing. A scenario giving both a data size and a link speed is asking you to calculate.

Meridian Freight must move 480 TB of archived shipment imagery from a regional depot to Amazon S3. The depot has a 200 Mbps internet connection shared with operations. The data centre closes in ten weeks.

The team has proposed copying it over the internet using the AWS CLI.

What should the solutions architect advise?

- ☐ A Do the arithmetic: at 200 Mbps the transfer exceeds the deadline, so use Snow Family devices
- ☐ B Proceed with the CLI, using multipart upload for parallelism
- ☐ C Use S3 Transfer Acceleration to speed up the upload
- ☐ D Use DataSync over the existing connection

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A Arithmetic, then Snow

200 Mbps is about 2 TB a day at perfection, so 480 TB is roughly 240 days. Ten weeks is 70. The link cannot do it.

C Transfer Acceleration

Improves throughput over long distances by using the AWS backbone sooner. It cannot exceed the depot's 200 Mbps.

B CLI with multipart

Parallelism uses the link more efficiently. It does not make the link faster, and the link is the constraint.

D DataSync

The best online tool available, and still bounded by the same 200 Mbps. Better software, same physics.

THE SIGNAL

Three options optimise the transfer and one questions whether it can finish. The deadline against the link speed is the question.

A media company must move 90 TB from an on-premises NFS share to Amazon FSx for Lustre, and then keep the two synchronised for three months while applications cut over in waves. The site has a 10 Gbps Direct Connect link. Integrity must be verified for every file. An engineer has proposed a nightly rsync job.

What should the solutions architect recommend?

- ☐ A AWS DataSync, with scheduled incremental tasks and built-in integrity validation
- ☐ B The nightly rsync job as proposed
- ☐ C Snow Family devices for the initial 90 TB, then rsync
- ☐ D S3 Transfer Acceleration into S3, then copy to FSx

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A** DataSync, scheduled incremental
90 TB over 10 Gbps is a day or two. DataSync then handles the three months of incremental sync, with validation and retries built in.
- B** Nightly rsync
Rebuilds DataSync badly. No integrity validation, no managed retries, no throttling, and somebody owns the script for three months.
- C** Snow then rsync
Ships a box for data that transfers in under two days over the existing link, then still needs the ongoing sync solved.
- D** Transfer Acceleration to S3
Adds a hop through S3 that the requirement does not need, and Transfer Acceleration does not target FSx.

THE SIGNAL

The arithmetic runs both ways. Here it rules Snow out, because 90 TB over 10 Gbps finishes long before a device could be shipped.

After migrating to AWS, 340 external partners must continue sending daily files. They currently connect to an on-premises SFTP server with credentials issued years ago. Renegotiating integration with each partner is not feasible.

The files must land in S3 for processing.

What should the solutions architect recommend?

- ☐ A AWS Transfer Family, providing a managed SFTP endpoint backed by S3
- ☐ B Run the SFTP server on EC2 in AWS
- ☐ C Ask partners to upload via an API Gateway endpoint
- ☐ D Use DataSync to collect files from each partner

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Transfer Family

Partners keep using SFTP exactly as they do now, files land directly in S3, and AWS operates the endpoint. No partner changes anything.

C API Gateway

Requires 340 partners to build new integrations. The scenario says renegotiating with each is not feasible.

B SFTP on EC2

Preserves the protocol and keeps you patching, scaling and securing an SFTP server, which is what Transfer Family removes.

D DataSync

Moves data between systems you control. It does not receive inbound uploads from third parties.

THE SIGNAL

This is not a migration question at all. It is a steady-state integration that happens to appear during a migration.

A company collects large video files from field teams in Australia, Brazil and South Africa, uploaded to an S3 bucket in eu-west-1. Uploads are slow and frequently fail partway. Teams in Ireland and the Netherlands report no problems.

Changing the bucket's Region is not possible for data residency reasons.

What should the solutions architect recommend?

- ☐ A Enable S3 Transfer Acceleration on the bucket, so distant uploads enter the AWS backbone at a nearby edge location
- ☐ B Create buckets in each Region and replicate to eu-west-1
- ☐ C Ship Snow Family devices to each field team
- ☐ D Use DataSync agents at each field location

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Transfer Acceleration

Distant clients are slow and nearby ones are fine, which is the exact profile. Uploads reach the AWS backbone at a nearby edge instead of crossing the internet.

C Snow devices

For bulk one-off movement, not continuous uploads from field teams. Shipping a device per upload is not a process.

B Regional buckets and replicate

Stores the data in Regions the residency requirement excludes, even briefly. It also multiplies buckets and transfer cost.

D DataSync agents

Suits a site with a file system to synchronise. Field teams uploading individual files are not that shape.

THE SIGNAL

The geography is the diagnosis. Nearby teams fine, distant teams failing, means the problem is the internet path, not the bucket.

A research institute must move 2 PB from a remote observatory with a 50 Mbps satellite link. The site closes in eight months. A junior architect has proposed AWS Snowmobile, having read about it in a study guide.

The organisation has used Snowball Edge devices before.

What should the solutions architect advise?

- ☐ A Snowmobile no longer exists; cluster multiple Snowball Edge Storage Optimized devices, each holding 210 TB
- ☐ B Order a Snowmobile as proposed
- ☐ C Transfer over the satellite link with DataSync bandwidth throttling
- ☐ D Compress the data and transfer over the satellite link

◆ PAUSE AND DECIDE

What should the solutions architect advise?

- A** Cluster Snowball Edge devices
Snowmobile was retired in 2024. Ten 210 TB devices cover 2 PB, and the organisation is an existing Snow customer so can still order them.
- B** Order a Snowmobile
It no longer exists. Study guides written before 2024 still describe it as the answer for petabyte-scale transfers.
- C** DataSync over satellite
50 Mbps is about 0.5 TB a day. 2 PB would take roughly eleven years. The tool is irrelevant at that ratio.
- D** Compress and transfer
Even at an implausible 50% ratio, five years instead of eleven. Compression does not bridge three orders of magnitude.

THE SIGNAL

Snowmobile is the most commonly cited dead answer in migration study material. Petabyte-scale is now clustered Snowball devices.

A company must move 700 TB from a data centre to S3. It has a 10 Gbps Direct Connect link currently running at about 20% utilisation, and a nine-week window. Security prefers data never leaves the private link.

The team is debating Snowball devices against an online transfer.

What should the solutions architect recommend?

- ☐ A DataSync over Direct Connect, throttled to leave headroom for production traffic
- ☐ B Snowball Edge devices, since 700 TB is large
- ☐ C S3 Transfer Acceleration over the internet
- ☐ D Split the data and use both Snowball and DataSync

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A

 DataSync over Direct Connect
80% of 10 Gbps is roughly 8 TB a day, so 700 TB takes about nine days against a nine-week window. It also stays on the private link.
- B

 Snowball devices
Chooses by data size rather than by arithmetic. Four devices, shipped and returned, to do what the existing link does in nine days.
- C

 Transfer Acceleration
Routes over the internet, which the security preference is specifically against. The private link is already there and faster.
- D

 Both
Two workstreams, two sets of tooling and reconciliation, to solve a problem one of them handles comfortably alone.

THE SIGNAL

700 TB sounds like a Snow scenario and is not. The link speed, not the data size, decides, and the private path is a bonus.

What to carry out of this module

- ◆ **Divide data by bandwidth before choosing anything.** Roughly, TB per day equals Gbps times ten, then discount for reality.
- ◆ **Snowmobile is retired and Snowcone is gone.** Snowball support ends in commercial Regions on 31 December 2026.
- ◆ **DataSync is the ongoing-sync answer**, with validation, retries, scheduling and throttling a script will not have.
- ◆ **Transfer Family is steady-state integration**, not migration. Transfer Acceleration helps distant clients only.

ON THE EXAM

The Snow Family remains examinable while being closed to new customers and dated for retirement. Answer with it; know the reality.